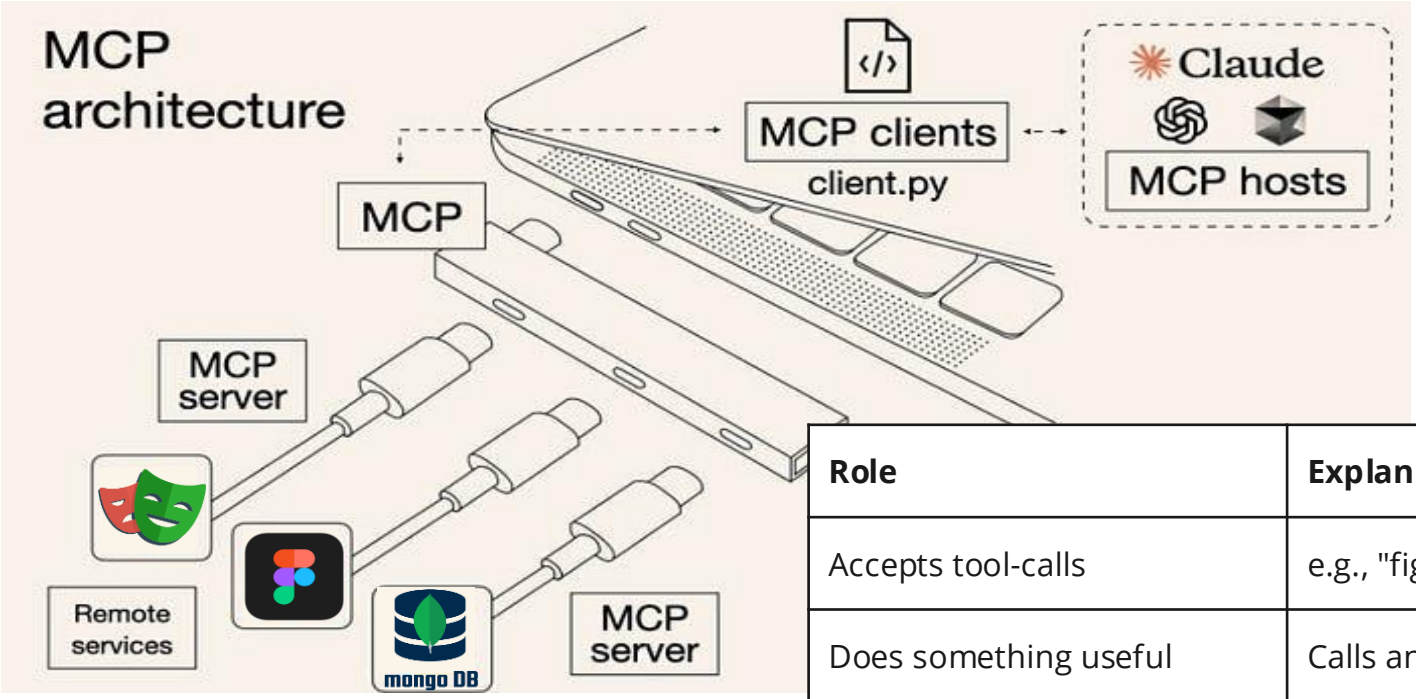
Build a working MCP Server of your own

PALO IT

tiny.cc/rni1101

Information Classification: CONFIDENTIAL (sensitive business information, the level of protection is dictated by legal agreements)

RECAP / What *is* an MCP Server really doing?



Role	Explanation
Accepts tool-calls	e.g., "figma.get_color"
Does something useful	Calls an API, reads a file, looks up a DB
Returns structured results	In JSON-RPC 2.0 format

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BUILD YOUR OWN MCP SERVER

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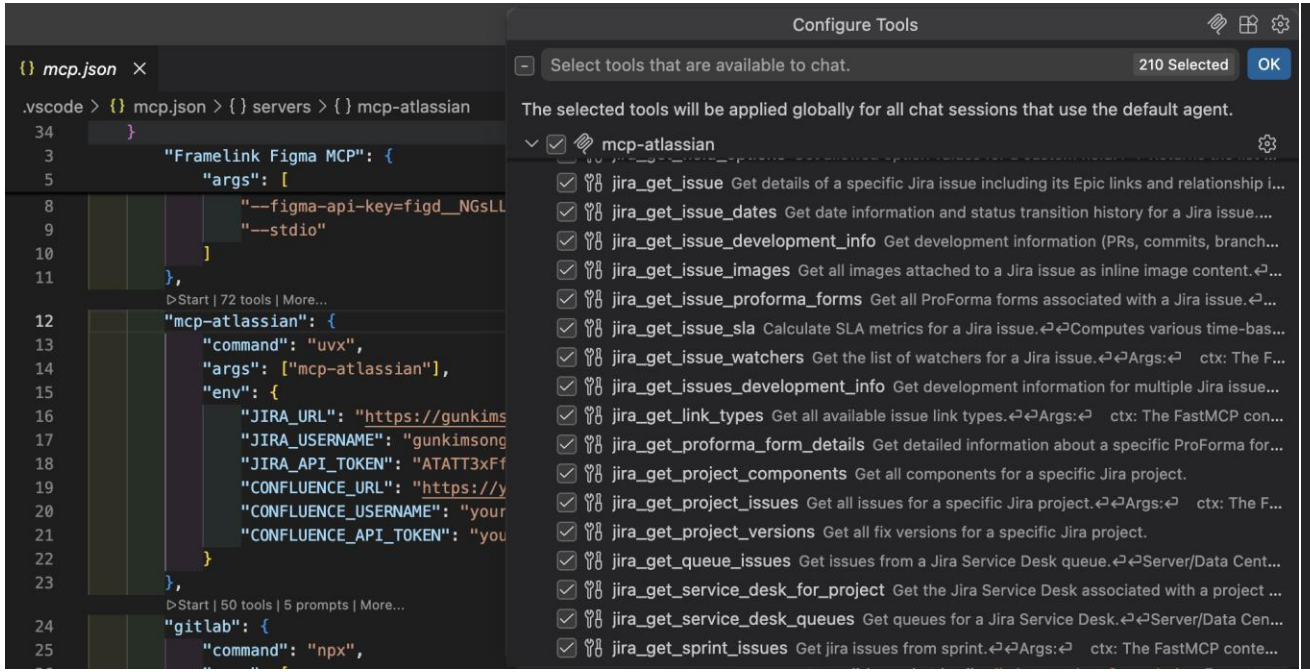

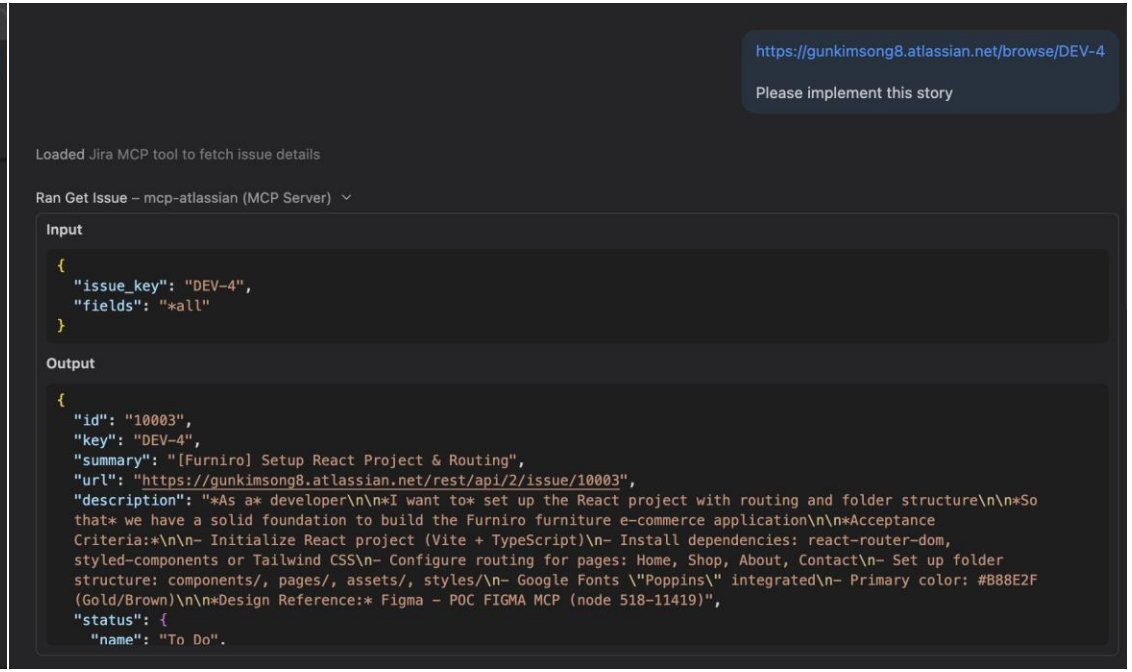

Accepts tool-calls

Does something useful and return In JSON-RPC 2.0 format

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🧩 MCP Server Config Types in IDEs

```
"servers": {
    ✓Running | Stop | Restart | More...
    "atlassian": {
        "type": "http",
        "url": "https://mcp.atlassian.com/v1/mcp"
    }
}
```

```
▷Start | 2 tools | More...
"Framelink Figma MCP": {
    "command": "npx",
    "args": [
        "-y",
        "figma-developer-mcp",
        "--figma-api-key=figd__NGsLLUIsiYylWrUFK4Jzk6Nh-0f8Nznkn8EYm
        "--stdio"
    ]
},
```

Config Type	Example	How it Works	Best Use
🌐 url (remote)	"url": "http://localhost:3000"	Connects directly to an already running MCP server via HTTP	Use in dev, cloud, or CI environments
🛠 npx or command (local)	"command": "npx mcp-mobile"	Spins up a temporary MCP server via CLI at runtime	Quick local development
🐳 docker	"docker": "mobile-next/mcp"	Launches a Docker container running the MCP server	Isolated dev/test setups

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AGENDA / OBJECTIVE

🧱 Part 1: Foundation (Concept + Mental Model)

- 🧠 Understand the message flow
  - How Copilot Agent communicates with your server
  - Key methods: tools/list, tools/call
  - Structure of a valid JSON-RPC response

🛠️ Part 2: Hands-on (Build + Run + Extend)

- 🧩 Design Your Tool Spec
  - Define tool name, description, and parameters
- 💻 Implement the Tool Logic
  - Call your service and format the response
- 🤖 Connect to Copilot
  - Setup .copilot/mcp.json
  - Test tool-call through chat prompt
- 🛠️ Debug & Validate
  - Use console logs, curl, or stdin to test edge cases
  - Handle errors gracefully in your respond() logic

By the end of this session, you'll have a working MCP Server and the confidence to build tools your Copilot can actually use.

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Part 1: Foundation

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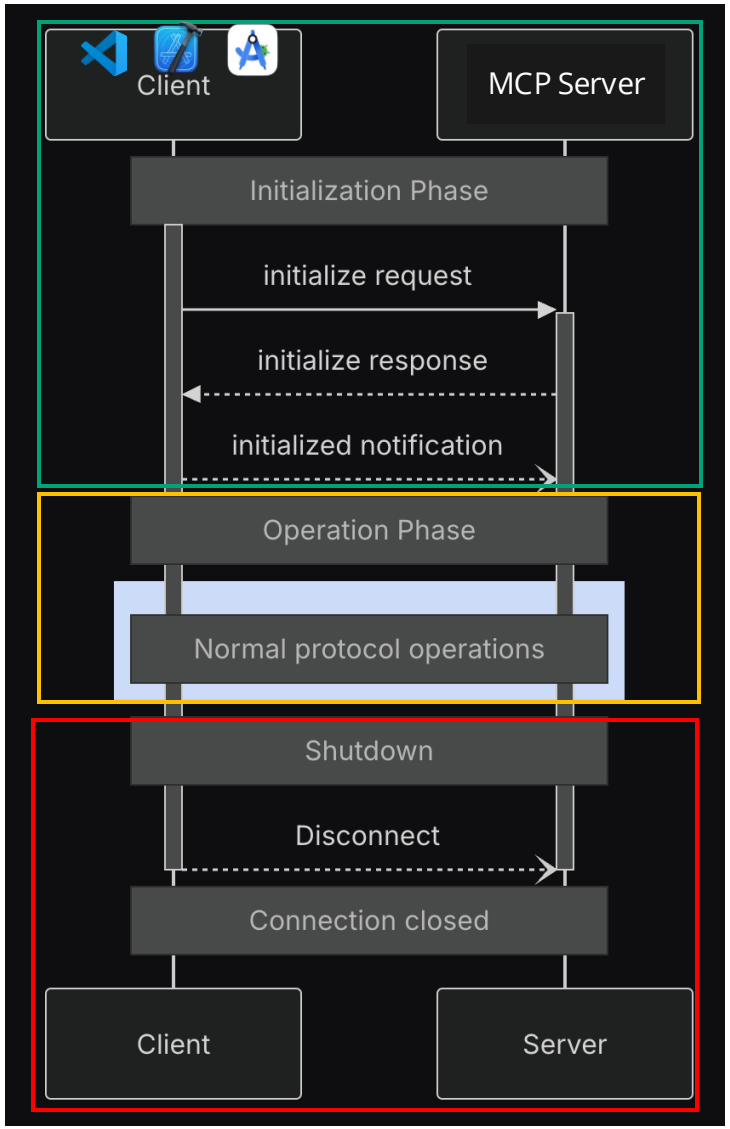
UNDERSTANDING THE ANATOMY OF MCP SERVER



Phase	Description	What Your MCP Server Must Do
Initialization	Agent connects and sends a initialize request	Respond with a valid JSON-RPC initialize response
Initialized	Agent confirms the connection is ready by sending initialized notification	The server MUST respond with its own capabilities and information:
Operation	Agent sends actual tool requests (tools/list, tools/call, etc.)	Handle JSON-RPC methods properly (e.g., tools/list, tools/call)
Shutdown	Agent sends shutdown request to gracefully close the session	No need actions.
Disconnect / Close	Agent closes the connection completely	Clean up if necessary (e.g., stop server, clear memory, etc.)

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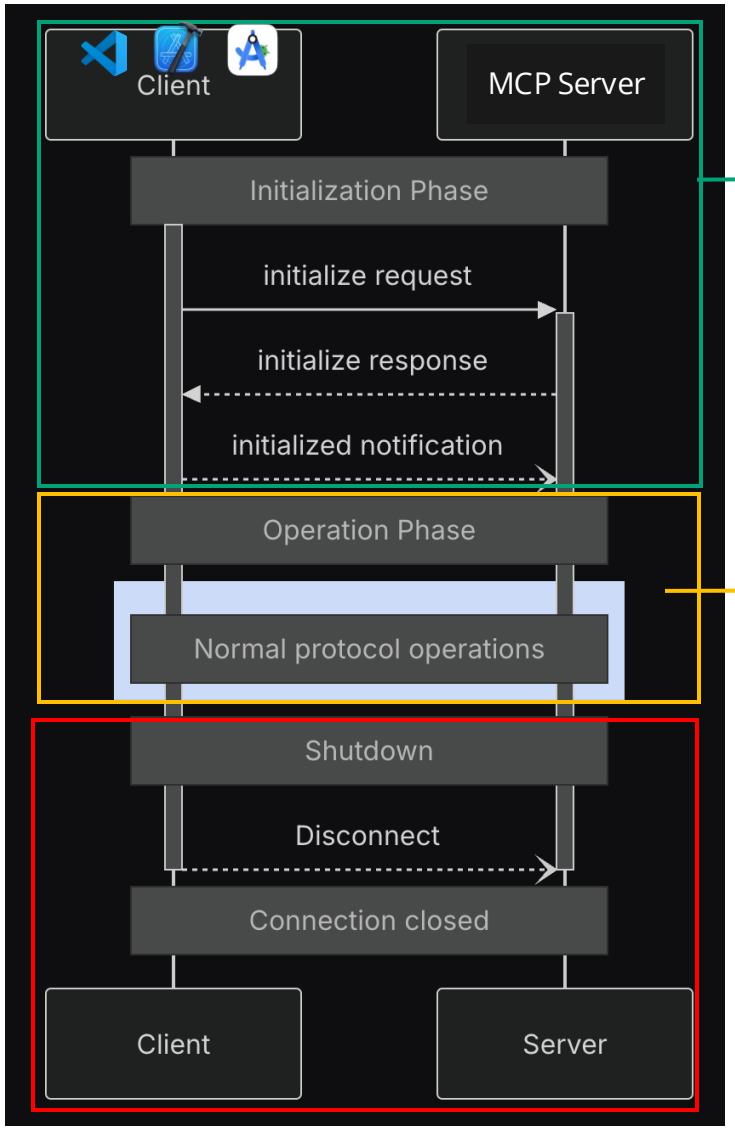
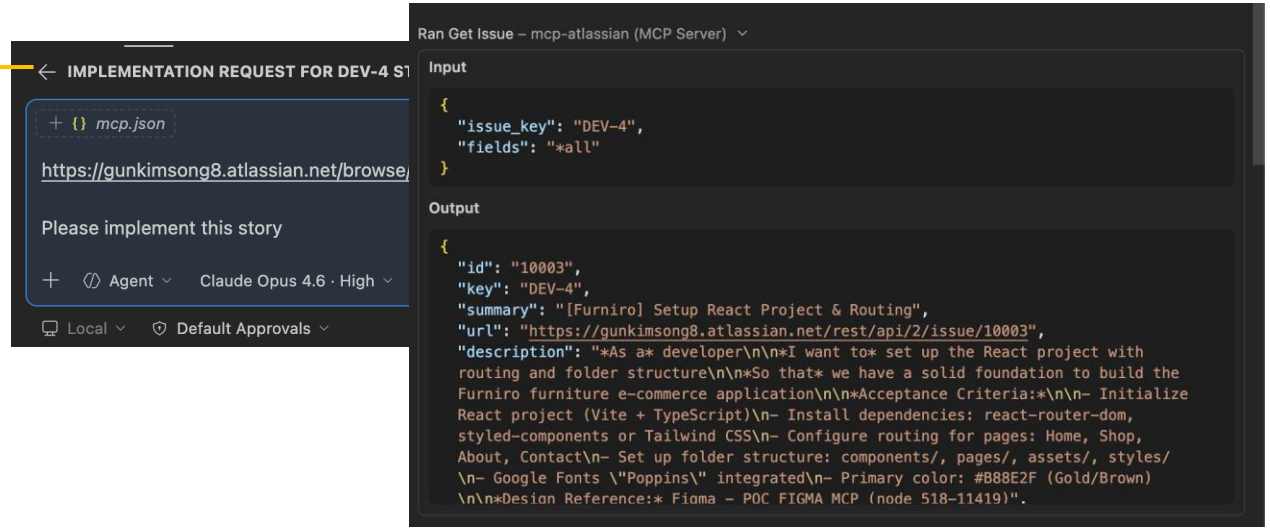
UNDERSTANDING THE ANATOMY OF MCP SERVER




```
e > {} mcp.json > ...
{
    "servers": {
        ▷Start | 31 tools | More...
        "atlassian": {
            "type": "http",
            "url": "https://mcp.atlassian.com/v1/mcp"
        }
    }
}
```



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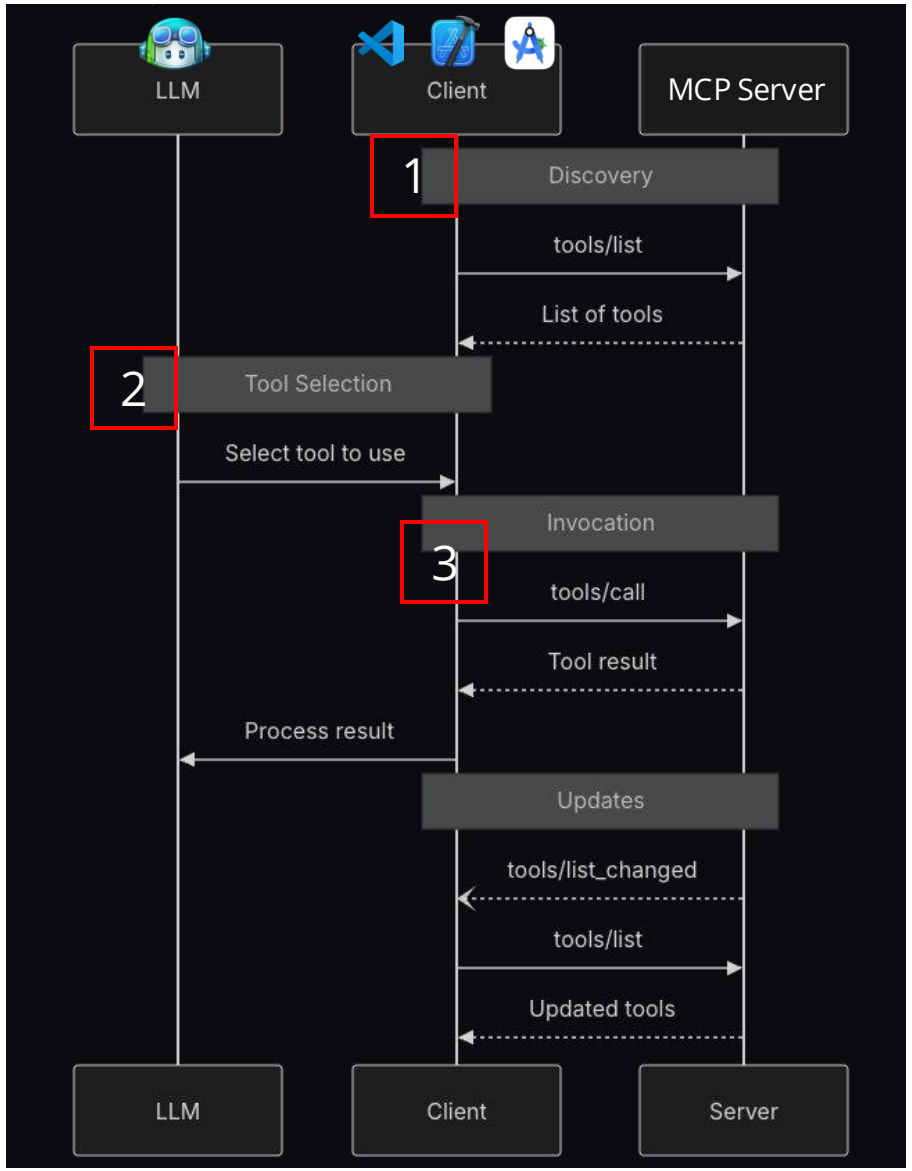
🧠 UNDERSTANDING THE OPERATION PHASE



🟣 1. Discovery – tools/list

- Client → Server: Requests a list of available tools
- Server → Client: Responds with the list of tool definitions (in JSON format)
- ✅ Your server needs to implement the tools/list method (or a route that supports it)

🎯 2. Tool Selection

- LLM (Copilot): Reads the tool metadata and decides which tool matches the user prompt
- Client → LLM: Sends selection request internally (not visible to server)

⚙️ 3. Invocation – tools/call

- Client → Server: Sends a JSON-RPC request to call the selected tool
- Server → Client: Responds with tool result
- ✅ Your MCP Server must respond to tools/call with the correct jsonrpc, id, and result

🔁 Updates – tools/list_changed

- Used when tools are dynamically added/removed
- Client → LLM: Notifies that tool list has changed
- LLM → Client → Server: Requests new tool list via tools/list again
- ✅ Optional: Only needed if your tool list changes at runtime

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STRUCTURE OF A VALID JSON-RPC REQUESTS

```
{
  jsonrpc: "2.0";
  id: string | number;
  method: string;
  params?: {
    [key: string]: unknown;
  };
}
```

Field	Type	Required	Description
jsonrpc	string	✅ Yes	Must be "2.0" to indicate JSON-RPC 2.0 protocol version
id	string / number / null	✅ Yes for request Optional for notification	Used to match responses with requests. If null, it's a notification
method	string	✅ Yes	Name of the method to be invoked
params	object / array	Optional	Parameters to be passed to the method. Named (object) or positional (array)

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STRUCTURE OF A VALID JSON-RPC RESPONSES

```
{
  jsonrpc: "2.0";
  id: string | number;
  result?: {
    [key: string]: unknown;
  }
  error?: {
    code: number;
    message: string;
    data?: unknown;
  }
}
```

Field	Type	Required	Description
jsonrpc	string	✅ Yes	Must be "2.0" to indicate JSON-RPC 2.0 protocol version
id	string / number / null	✅ Yes for request Optional for notification	Used to match responses with requests. If null, it's a notification
method	string	✅ Yes	Name of the method to be invoked
result	any	✅ Yes (in response)	Result of a successful method invocation
error	object	✅ Yes (in error response)	Error info: must contain code, message, optional data

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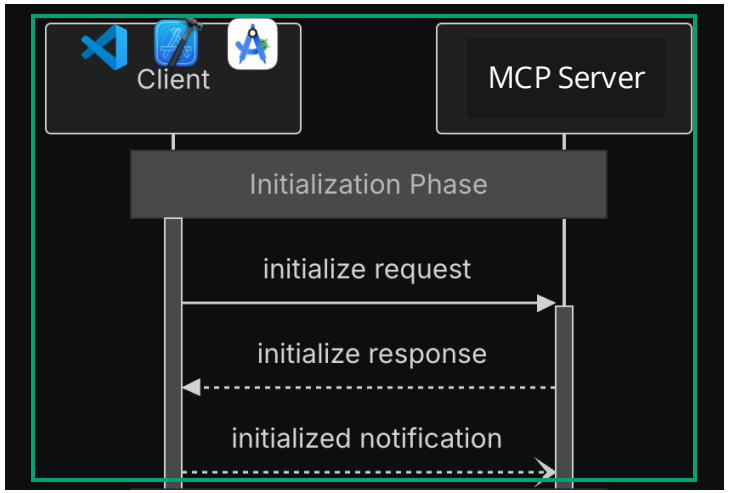
UNDERSTANDING THE ANATOMY OF MCP SERVER




```
e > {} mcp.json > ...
{
    "servers": {
        ▷Start | 31 tools | More...
        "atlassian": {
            "type": "http",
            "url": "https://mcp.atlassian.com/v1/mcp"
        }
    }
}
```

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "method": "initialize",
  "params": {
    "protocolVersion": "2025-11-25",
    "capabilities": {
      "roots": {
        "listChanged": true
      },
      "sampling": {},
      "elicitation": {
        "form": {},
        "url": {}
      },
      "tasks": {
        "list": {},
        "cancel": {},
        "requests": {
          "sampling": {
            "createMessage": {}
          },
          "elicitation": {
            "create": {}
          }
        }
      },
      "extensions": {
        "io.modelcontextprotocol/ui": {
          "mimeTypes": [
            "text/html;profile=mcp-app"
          ]
        }
      }
    },
    "clientInfo": {
      "name": "Visual Studio Code",
      "version": "1.113.0"
    }
  }
}
```

Field	Type	Required
jsonrpc	string	✅ Yes
id	string / number / null	✅ Yes for re Optional for notification
method	string	✅ Yes
params	object / array	Optional

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🧠 UNDERSTANDING THE ANATOMY OF MCP SERVER

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "result": {
    "protocolVersion": "2024-11-05",
    "capabilities": {
      "tools": {
        "listChanged": false
      }
    },
    "serverInfo": {
      "name": "weather-service",
      "version": "1.0.0"
    }
  }
}
```

Field	Type	Required
jsonrpc	string	✅ Yes
id	string / number / null	✅ Yes for request Optional for notification
method	string	✅ Yes
result	any	✅ Yes (in response)
error	object	✅ Yes (in error response)

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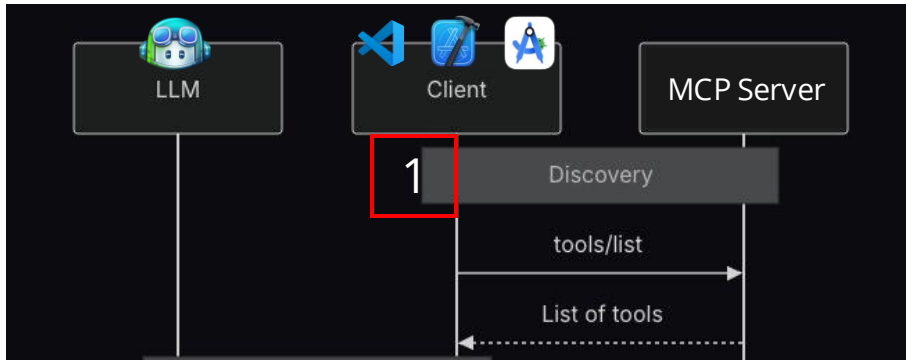

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- Client → Server: Requests a list of available tools
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- ✅ Your server needs to implement the tools/list method (or a route that supports it)

```
🔧 MCP TOOLS/LIST: Returning 3 tools
{
  "jsonrpc": "2.0",
  "id": 2,
  "result": {
    "tools": [
      {
        "name": "weather_get_current",
        "description": "Get current weather information for a specified city",
        "inputSchema": {
          "type": "object",
          "properties": {
            "city": {
              "type": "string",
              "description": "The name of the city to get weather for (e.g., 'bangkok', 'london', 'tokyo')"
            }
          },
          "required": [
            "city"
          ]
        }
      },
```

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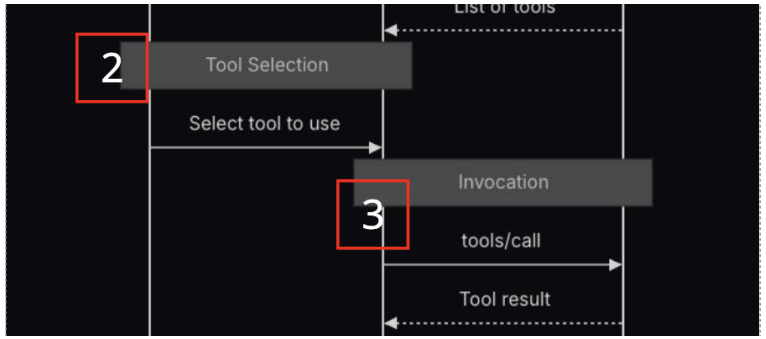
STRUCTURE OF A VALID JSON-RPC RESPONSE

To invoke a tool, clients send a tools/call request:



3. Invocation – tools/call

- Client → Server: Sends a JSON-RPC request to call the selected tool
- Server → Client: Responds with tool result
- ✅ Your MCP Server must respond to tools/call with the correctjsonrpc, id, and result

```
{
  "jsonrpc": "2.0",
  "id": 2,
  "method": "tools/call",
  "params": {
    "name": "get_weather",
    "arguments": {
      "location": "New York"
    }
  }
}
```

RESPONSE

```
{
  "jsonrpc": "2.0",
  "id": 2,
  "result": {
    "content": [
      {
        "type": "text",
        "text": "Current weather in New York:\nTemperature: 72°F\nConditions: Partly cloudy"
      }
    ],
    "isError": false
  }
}
```

REQUEST

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🛠️ Part 2: Hands-on (Build + Run + Extend)

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FEEDBACK
Q&A

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